INCH – POUND

MIL-STD-1861
23 August 1989

MILITARY STANDARD

ELECTRICAL AND ELECTRONIC ASSEMBLIES, BOARDS, CARDS, AND ASSOCIATED HARDWARE,

SELECTION AND USE OF

AMSC N/A

FSC 5998

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FOREWORD

1. This military standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. This standard provides a list of electrical and electronic assemblies, boards, cards, and associated hardware for use in the design of military equipment.

4. The application information and performance characteristics contained in this standard are offered for guidance and are not to be considered as mandatory.

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1. SCOPE

1.1 Scope. This standard establishes the criteria for the design and treatment of printed wiring boards, assemblies and associated hardware.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-13949	-	Plastic Sheet, Laminated, Copper Clad (For Printed Wiring).
MIL-C-28809	-	Circuit Card Assemblies, Rigid, Flexible, and Rigid-Flex.
MIL-A-28870	-	Assemblies, Electrical Backplane, Printed Wiring, General Specification for.
MIL-I-46058	-	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies).
MIL-P-46843	-	Printed Wiring Assemblies.
MIL-P-50884	-	Printed Wiring, Flexible, and Rigid-Flex.
MIL-P-55110	-	Printed Wiring Boards.

STANDARDS

MILITARY

MIL-STD-275	-	Printed Wiring for Electronic Equipment.
MIL-STD-2118	-	Flexible and Rigid-Flex Wiring for Electronic Equipment, Design Requirements for.
MIL-STD-2119	-	Design Requirements for Printed Wiring Electrical Backplane Assemblies.

HANDBOOKS

FEDERAL

H-6	-	Alphabetic Index of Names.
H2-2	-	Cataloging Handbook.

(Copies of Cataloging Handbooks H2-2 and H6 are available from the Commander, Defense Logistics Services Center, Battle Creek, MI 49017-3084. Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Naval Publications and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

83023	-	Extractor, Electrical Card, Plastic.
84006	-	Electrical Card Holders.
84101	-	Electrical Card Holders.

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84103 - Electrical Card Holders, Retainers.
84168 - Electrical Card Holders, Retainers.
84191 - Extractor, Electrical Card, Metal.
85020 - Holder, Electrical Card.
85021 - Holder, Electrical Card.
85033 - Holder, Electrical Card.
85034 - Holder, Electrical Card.
85036 - Spacer, Electrical Card Holder.
85067 - Electrical Card Holders.
85068 - Holder, Electrical Card.
85075 - Support, Electrical Card Holder, Bar Mounting.
85076 - Holder, Electrical Card.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS (IPC)

IPC-DW-425/11 - Design and End Product Requirements for Discrete Wiring Boards/Plated-Through Hole Connection.

(Application for copies should be addressed to the Institute for Interconnecting and Packaging Electronic Circuits, 7380 North Lincoln Avenue, Lincolnwood, IL 60646.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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4. GENERAL REQUIREMENTS

4.1 Purpose. The purpose of this standard is to:

a. Provide the designers of new equipment with a selection of standard items for military applications.

b. Control and minimize the variety of items used in military equipment in order to facilitate logistic support of the equipment in the field.

c. Outline criteria pertaining to the use of items for design purposes for new equipment.

4.2 Conflict of requirements. In the event of conflict between the technical requirements of items described in this standard and the applicable specification or drawing, the text of this document shall govern.

4.3 Criteria for inclusion in this standard. The criteria for inclusion in this standard are as follows:

a. The item(s) are classified in FSC 5998.

b. The item(s) shall be the best type available for general use in military equipment.

c. The item(s) are covered by military specifications or drawings.

d. The item(s) shall be available from at least one source.

e. The item(s) shall be used for design purposes in new equipment.

4.4 Order of preference. The order of preference shall be as follows:

a. Military specification parts.

b. Military drawing parts.

c. Other parts.

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5. DETAILED REQUIREMENTS

5.1 Rigid printed wiring and printed wiring boards. Rigid printed wiring and printed wiring boards for single-sided, double-sided, and multilayer printed wiring shall conform to MIL-STD-275 and MIL-P-55110. The materials used for single-sided, double-sided, and multilayer printed wiring boards shall conform to MIL-P-13949.

5.2 Printed wiring board size. Whenever cost and technical requirements permit, preferred rigid printed wiring board sizes should be used. These board sizes will facilitate the development and use of standardized insertion end extraction tools. The preferred board sizes, extractor hole size and location are shown on figure 1.

5.3 Rigid printed wiring assemblies. Rigid printed wiring assemblies consisting of rigid printed wiring boards on which separately manufactured parts have been added shall conform to MIL-C-28809. For Army missile weapon systems, MIL-P-46843 shall also apply.

5.4 Conformal coating. When conformal coating is required, rigid printed wiring assemblies shall be conformally coated with a material which conforms to MIL-I-46058.

5.5 Flexible and rigid-flex wiring. Flexible and rigid-flex printed wiring shall conform to MIL-P-50884 and shall be designed in accordance with MIL-STD-2118.

5.6 Discrete wiring boards. Discrete wiring boards with plated-through holes shall be in accordance with IPC-DW-425/11. Discrete wiring boards shall not be used for space application.

5.7 Backplane assemblies, printed wiring. Electrical backplane printed wiring assemblies shall conform to MIL-A-28870 and shall be designed in accordance with MIL-STD-2119.

5.8 Associated hardware.

5.8.1 Card guides. Card guides shall be as specified in tables I through V for metals, and tables VI through IX for plastics.

5.8.2 Extractors/ejectors. Extractors/ejectors shall be as specified in tables X and XI.

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TABLE I. Electrical card holders, metal, any slot width.

Overall length (inches)	Mounting method	Drawing part number 84103-
3.0	2 holes	01
3.2	"	02
3.7	"	03
4.0	"	04
5.0	"	05
5.5	"	06
6.0	"	07

TABLE II. Electrical card holders, metal, slot width .0312 inch.

Overall length (inches)	Mounting method	Drawing part number 85020-
2.0	2 holes	002
2.5	"	026
3.0	"	050
3.5	"	074
4.0	"	098
4.5	"	122
5.0	"	146
5.5	"	170
6.0	"	194

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TABLE III. Electrical card holders, metal, slot width .0625 inch.

Overall length (inches)	Mounting method	Drawing part number
2.0	2 holes	85020-005
2.5	"	85034-01
2.50	"	85020-029
2.50	"	85021-002
3.0	2 holes	84168-002
3.0	"	85020-053
3.0	"	85021-017
3.25	3 holes	85034-03
3.5	2 holes	84168-010
3.5	"	85020-077
3.5	"	85021-032
4.0	2 holes	84168-018
4.0	"	85020-101
4.0	"	85021-047
4.25	4 holes	85034-05
4.5	2 holes	84168-026
4.5	2 holes	85020-125
4.5	3 holes	85033-001
5.0	3 holes	84168-034
5.0	2 holes	85020-149
5.0	2 holes	85021-062
5.25	5 holes	85034-07
5.5	3 holes	84168-042
5.5	2 holes	85020-173
6.0	3 holes	84168-050
6.0	2 holes	85020-197
6.0	2 holes	85021-077
6.0	4 holes	85033-017
6.25	6 holes	85034-09
7.0	3 holes	84168-058
7.5	3 holes	84168-066
8.25	8 holes	85034-11
8.5	3 holes	84168-074

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TABLE IV. Electrical card holders, metal, slot width .0937 inch.

Overall length (inches)	Mounting method	Drawing part number
1.5	2 holes	85033-135
2.0	"	85020-008
2.25	"	85034-02
2.50	"	85020-032
2.50	"	85021-005
3.00	2 holes	85020-056
3.00	"	85021-020
3.00	"	84168-004
3.00	3 holes	85033-147
3.25	3 holes	85034-04
3.5	2 holes	85020-080
3.5	"	85021-035
3.5	"	84168-012
4.0	2 holes	85020-104
4.0	"	85021-050
4.0	"	84168-020
4.25	4 holes	85034-06
4.5	2 holes	84168-028
4.5	2 holes	85020-128
4.5	3 holes	85033-005
5.0	2 holes	85020-152
5.0	2 holes	85021-065
5.0	3 holes	84168-036
5.25	5 holes	85034-08
5.5	2 holes	85020-176
5.5	3 holes	84168-044
6.0	4 holes	85033-020
6.0	2 holes	85020-200
6.0	2 holes	85021-080
6.0	3 holes	84168-052
6.25	6 holes	85034-10
7.0	3 holes	84168-060
7.5	3 holes	84168-068
8.25	8 holes	85034-12
8.5	3 holes	84168-076

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TABLE V. Electrical card holders, metal, slot width .125 inch.

Overall length (inches)	Mounting method	Drawing part number
1.5	2 holes	85033-185
2.0	"	85033-189
2.5	"	85033-031
3.0	3 holes	85033-035
3.5	"	85033-039
4.0	"	85033-043
4.5	"	85033-047
5.0	4 holes	85033-051
5.5	"	85033-054
6.0	"	85033-057

TABLE VI. Electrical card holders, metal, slot width .063 inch.

Overall length (inches)	Mounting method	Drawing part number
2.0	2 holes	85068-01
3.5	"	85068-02
5.0	"	85068-03

TABLE VII. Electrical card holders, plastic, slot width .078 inch.

Overall length (inches)	Mounting method	Drawing part number
2.5	2 snaps	84006-01
3.0	"	84006-02
3.5	"	84006-03
4.0	"	84006-04
4.5	"	84006-05
5.0	"	84006-06
5.5	"	84006-07
6.0	"	84006-08
6.5	"	84006-09
7.0	"	84006-10
7.5	"	84006-11
8.0	"	84006-12
8.5	3 snaps	84006-13

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TABLE VII. Electrical card holders, plastic, slot width .078 inch - Continued.

Overall length (inches)	Mounting method	Drawing part number
9.0	3 snaps	84006-14
9.5	"	84006-15
10.0	"	84006-16
10.5	"	84006-17
11.0	"	84006-18
11.5	"	84006-19
12.0	"	84006-20
12.5	"	84006-21
13.0	"	84006-22
13.5	"	84006-23
14.0	"	84006-24

TABLE VIII. Electrical card holders, plastic, slot width .080 inch.

Overall length (inches)	Mounting method	Drawing part number
2.50	2 snaps	84101-01
4.50	"	84101-02
6.00	"	84101-03
8.00	3 snaps	84101-04

TABLE IX. Electrical card holders, plastic, slot width .050 to .100 inch.

Overall length (inches)	Mounting method	Drawing part number
2.50	2 snaps	85067-09
2.50	"	84006-08
3.50	"	84006-07
4.125	"	84006-06
4.125	"	84006-05
5.50	"	84006-04
6.375	"	84006-03
6.875	"	84006-02
7.00	3 snaps	84006-01

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TABLE X. Extractor, electrical card, plastic.

Material	Slot inch	Drawing part number
Nylon	.075	83023-01*
Nylon	.075	83023-02*
Nylon	.075	83023-03*
Nylon	.075	83023-04*
Nylon	.105	83023-05*
Nylon	.105	83023-06*
Nylon	.105	83023-07*
Nylon	.100	83023-08*
Glass filled nylon	.075	83023-09*
Glass filled nylon	.140	83023-10*
Glass filled nylon	.075	83023-11*
Glass filled nylon	.075	83023-12*
Nylon	.070	83023-13*
Glass filled polycarbonate	.075	83023-14*
Nylon	.070	83023-15*
Nylon	.100	83023-16*
Glass filled polycarbonate	.140	83023-17*
Nylon	.070	83023-18*
Nylon	.075	83023-19*
Nylon	.100	83023-20*
Nylon	.140	83023-21*

* Indicates color.

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TABLE XI. Extractor, electrical card, aluminum.

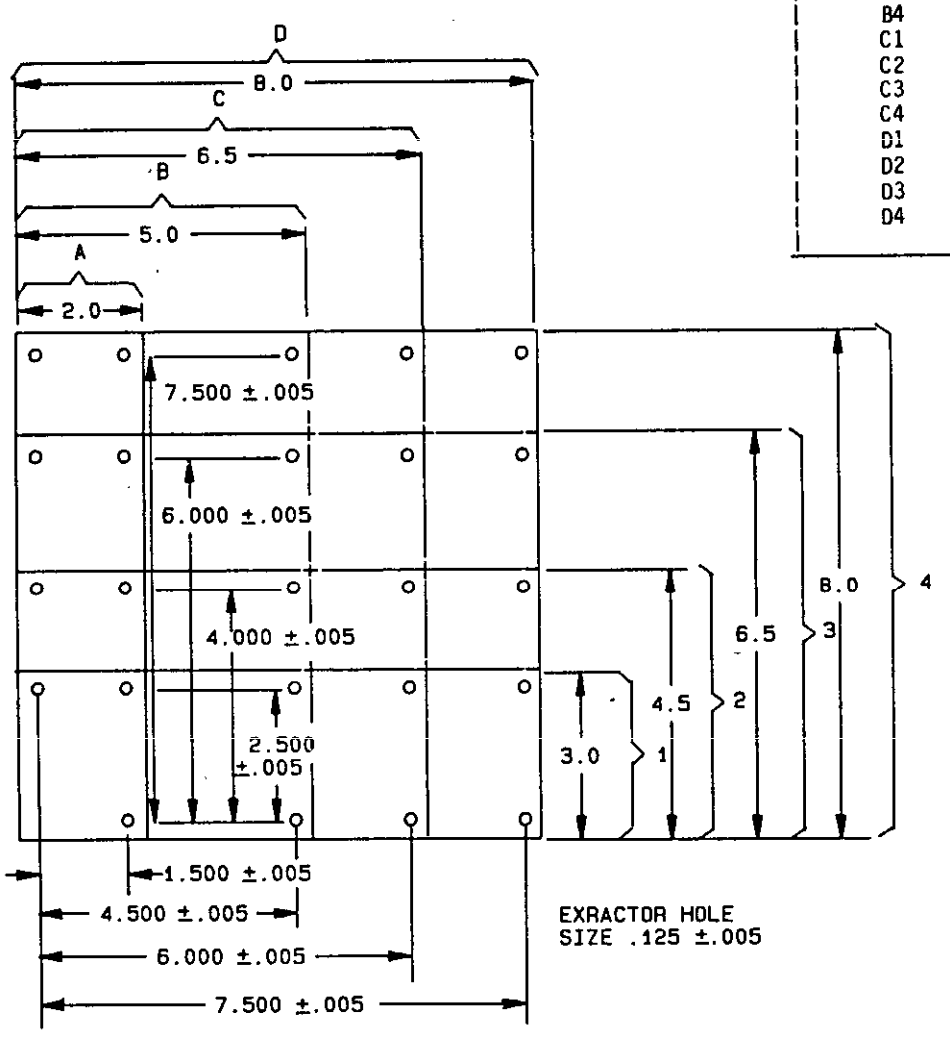
Slot inch	Drawing part number	Configuration
.040	84191-01*	Angled handle, no lock tab
.040	84191-02*	
.075	84191-03*	
.075	84191-04*	
.105	84191-05*	
.105	84191-06*	
.142	84191-07*	
.142	84191-08*	
.040	84191-09*	Angled handle, with lock tab
.040	84191-10*	
.075	84191-11*	
.075	84191-12*	
.105	84191-13*	
.105	84191-14*	
.142	84191-15*	
.142	84191-16*	
.040	84191-17*	Straight handle, with lock tab
.040	84191-18*	
.075	84191-19*	
.075	84191-20*	
.105	84191-21*	
.105	84191-22*	
.142	84191-23*	
.142	84191-24*	
.040	84191-25*	Straight handle, no lock tab
.040	84191-26*	
.075	84191-27*	
.075	84191-28*	
.105	84191-29*	
.105	84191-30*	
.142	84191-31*	
.142	84191-32*	

* Indicates finish. The finish options are black anodize "B", chemical film "C" and chemical film with the electronic sensitive device type "E". The "E" finish is preferred for Air Force use.

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Preferred thicknesses	
Thickness	Tolerance
.062	±.001
.093	±.0015
.125	±.002

Board number	Printed wiring board size in nominal ±.015
A1	2.0 x 3.0
A2	2.0 x 4.5
A3	2.0 x 6.5
A4	2.0 x 8.0
B1	5.0 x 3.0
B2	5.0 x 4.5
B3	5.0 x 6.5
B4	5.0 x 8.0
C1	6.5 x 3.0
C2	6.5 x 4.5
C3	6.5 x 6.5
C4	6.5 x 8.0
D1	8.0 x 3.0
D2	8.0 x 4.5
D3	8.0 x 6.5
D4	8.0 x 8.0



Inches	mm
.001	0.03
.0015	0.038
.002	0.07
.005	0.13
.015	0.38
.062	1.57
.093	2.36
.125	3.18
1.500	38.100
2.0	51.
2.500	63.50
3.0	76.
4.000	101.60
4.500	114.30
5.0	127.
6.000	152.40
6.5	165.
7.500	190.50
8.0	203.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only.

FIGURE 1. Preferred printed wiring board sizes.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Documents conforming to the requirements of this standard are intended for the use as military standardization documents and are listed in the DODISS. The purpose of this standard is to standardize the preparation of military standards, handbooks, drawings, and bulletins to ensure the inclusion of data and descriptions essential to the selection and application of items and processes, and to aid in the use and analysis of DoD standardization documents.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - ME, MI, SL
Air Force - 11, 17, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-1861	Electrical and Electronic Assemblies, Boards, Cards and

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
Associated Hardware	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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